11. Sequences and Series

11.4. Comparison Tests for Series

11.4 Learning Objectives
- Determine when it is appropriate to use a comparison test
- Determine an appropriate series to compare a given series to
- Know when and how to use direct comparison and limit comparison tests.

11.4.1. Direct Comparison Test

Example 11.4.1

Determine whether the series $\sum_{n=1}^{\infty} \frac{1}{n-1}$ converges or diverges.

This series is extremely similar to $\sum_{n=2}^{\infty} \frac{1}{n}$, which we know diverges. In fact, thinking about graph transformations, $\frac{1}{x-1}$ is just $\frac{1}{x}$ shifted one space to the right.

Because $\sum_{n=2}^{\infty} \frac{1}{n}$ diverges, we expect that $\sum_{n=1}^{\infty} \frac{1}{n-1}$ also diverges, since the only difference is the -1 in the denominator... let's look at a picture

todo: plots

We can observe that $\frac{1}{x} < \frac{1}{x-1}$ for all $x > 0$, so for all $n \geq 1$ we have

$$\begin{aligned} \frac{1}{n} &< \frac{1}{n-1} \\ \sum_{n=2}^{\infty} \frac{1}{n} &\leq \sum_{n=2}^{\infty} \frac{1}{n-1} \\ \infty &\leq \sum_{n=2}^{\infty} \frac{1}{n-1} \end{aligned}$$

so $\sum_{n=2}^{\infty} \frac{1}{n-1}$ diverges since $\sum_{n=2}^{\infty} \frac{1}{n}$ diverges.

Theorem 11.4.1 The Direct Comparison Test (Comparison Test)

Suppose that $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are series with positive terms.

(i) If $\sum_{n=1}^{\infty} b_n$ is convergent and $a_n \leq b_n$ for all $n \geq N$, then $\sum_{n=1}^{\infty} a_n$ converges.

(ii) If $\sum_{n=1}^{\infty} b_n$ is divergent and $a_n \geq b_n$ for all $n \geq N$, then $\sum_{n=1}^{\infty} a_n$ diverges.

Note

In this theorem, we have a series $\sum_{n=1}^{\infty} a_n$ with positive terms that we wish to talk about the convergence or divergence of. We make an educated guess on if it converges or diverges, and find

another series $\sum_{n=1}^{\infty} b_n$ with positive terms either above or below the original, and hope that other series is easier to analyze the convergence of.

Note

The $n \geq N$ part of all these series convergence tests can be read as "eventually the series behaves this way..."

In the above theorem, it only matters that *eventually* we have $a_n \leq b_n$ (or $a_n \geq b_n$). The first few terms before N can only contribute a finite amount to the series, and don't end up mattering in the long run of the sum.

todo: plots

Example 11.4.2

Does the series $\sum_{n=1}^{\infty} \frac{2^n}{n+6^n}$ converge or diverge?

Solution

Let $a_n = \frac{2^n}{n+6^n}$. Doing a quick check of the divergence test (exercise: check the divergence test is inconclusive yourself, you will need h'Hosptial's rule and to review exponent rules and exponential derivatives!) we see that $\lim_{n\to\infty} a_n = 0$, so the series does not obviously diverge. Then we have to use another one of the tests.

Now we see that this series is very similar to $\sum_{n=1}^{\infty} \frac{2^n}{6^n}$ except it differes by the $+n$ in the denominator. Importantly,

$$\sum_{n=1}^{\infty} \frac{2^n}{6^n} = \sum_{n=1}^{\infty} \left(\frac{1}{3}\right)^n$$

converges by the Geometric Series test, since $r = \frac{1}{3} < 1$.

Now, we have to check the rest of the conditions of the comparison test: all the terms need to be positive and for $n \geq 1$ we have that both $\frac{2^n}{n+6^n}$ and $\frac{2^n}{6^n}$ are positive.

Lastly we need to show that the series we are comparing to has terms, $b_n \geq a_n$, where a_n are the terms in the original. Notice that if we want

$$\begin{aligned} \frac{2^n}{n+6^n} &< \frac{2^n}{6^n} \\ \text{then we must also have } \quad \frac{1}{n+6^n} &< \frac{1}{6^n} \\ \text{in which case we also would have } \quad n+6^n &> 6^n \\ \text{which is the case if } \quad n &> 0. \end{aligned}$$

Fortunately this is true, we have $n \geq 1 > 0$ for all terms in this series ($N = 0$).

So, **by the comparison test** the series $\sum_{n=1}^{\infty} \frac{2^n}{n+6^n}$ converges.

Note

- We don't have to check the divergence test is inconclusive for all of these, but if the divergence test works and shows something is divergent, it is a really quick and easy solution. Essentially, we don't want to miss any easy ways out in this process so it is good to at least think about before continuing on with the problem.
- A complete solution to determining a series converges or diverges has the following "rubric":
  - Did the problem solver identify the right test to use?
  - Did the problem solver check all the hypotheses for the test, showing enough detail for each?
  - Did the problem solver write a conclusion about the convergence of the series including a citation stating which test they used?
- Note these problems need more writing than a lot of other math up to this point, you should have a few *complete* sentences in a solution.

Example 11.4.3

Does the series $\sum_{n=1}^{\infty} \frac{\cos^2(n)}{1+n^5}$ converge or diverge?

Solution

With a quick limit, the divergence test is inconclusive:

$$\lim_{n\to\infty} \frac{\cos^2(n)}{1+n^5} \text{ behaves like a finite number divided by a very large number so } = 0$$

Now we notice that this series is similar to $\sum_{n=1}^{\infty} \frac{1}{n^5}$, which converges by the p-series test. We just have to relate the original series to this one

$$a_n = \frac{\cos^2(n)}{1+n^5} \leq \frac{1}{1+n^5} < \frac{1}{n^5} = b_n$$

(in the first step above, the denominator got bigger going from $\cos^2(n)$ to 1, then in the second step the denominator got smaller which made the entire fraction larger)

Lastly, for direct comparison test to work, the terms in both series must be positive. Notice that $\frac{1}{n^5}$ is positive for $n \geq 1$ and $\cos^2(n) \geq 0$ since it is a number between -1 and 1 that is squared.

Then, by the direct comparison test, $\sum_{n=1}^{\infty} \frac{\cos^2(n)}{1+n^5}$ also converges.

Warning

Can we apply the Direct Comparison Test to the series $\sum_{n=2}^{\infty} \frac{1}{n^2-1}$?

This series is similar to $\sum_{n=2}^{\infty} \frac{1}{n^2}$, which converges by the p-series test, so we would expect the original series to converge. However, since

$$n^2 - 1 < n^2 \quad \text{for} \quad n \geq 2$$

$$\text{we have} \quad \frac{1}{n^2 - 1} > \frac{1}{n^2}$$

and so our convergent series we want to compare to is below the original! So, the direct comparison test does not work.

We will build up a tool to use on these types of series in the next part of 11.4.

11.4a Section Summary:

- We identified when to use direct comparison test and how to find a series to compare to.
- We applied direct comparison test.